Sida Ye

48 Creighton St., Cambridge — Tel: +1 857-829-7969
MA 02140, USA — Email: siye2@bwh.harvard.edu / yesida.haumun@gmail.com

WORK EXPERIENCES

Harvard Medical School/Brigham and Women's Hospital/HHMI — Nov 2025 - Present
Postdoctoral scientist (Advisor: Prof. Matthew K. Waldor*)* — Boston, U.S.

University of Massachusetts Boston — June 2025 - Oct 2025
Postdoctoral fellow (Advisors: Prof. Kourosh Zarringhalam & Prof. Manoj T. Duraisingh) — Boston, U.S.

EDUCATION

University of Massachusetts Boston/Harvard T.H. Chan School of Public Health — Sep 2021- May 2025
Ph.D. in Computational Sciences — Boston, U.S.
Advisors: Prof. Kourosh Zarringhalam & Prof. Manoj T. Duraisingh

Peking University, School of Life Sciences — 2014 - 2020
M.S., Botany and Molecular Biology — Beijing, China

University of Cambridge, Sainsbury Laboratory — Sep 2017 - Dec 2017
Visiting student — Cambridge, U.K.

Huazhong Agricultural University, Zhidong Zhang Class — 2010 - 2014
B.S., Plant sciences and technology — Wuhan, China

PUBLICATIONS

- Elsworth B#, **Ye S#**, Dass S#, Tennessen JA, Sultana Q, Thommen BT, Paul AS, Kanjee U, Grüring C, Ferreira MU, Gubbels MJ, Zarringhalam K, Duraisingh MT. The essential genome of *Plasmodium knowlesi* reveals determinants of antimalarial susceptibility*.* **Science**. 2025;387(6734):eadq6241. PMID:39913579 (#co-first author).
- Keroack CD, Elsworth B, Tennessen JA, Paul AS, Hua R, Ramirez-Ramirez L, **Ye S**, Moreira CK, Meyers MJ, Zarringhalam K, Duraisingh MT. Comparative chemical genomics in Babesia species identifies the alkaline phosphatase PhoD as a determinant of antiparasitic resistance. **Proceedings of the National Academy of Sciences**. 2024 Feb 27;121(9):e2312987121. PMID: 38377214.
- Zarringhalam K, **Ye S**, Lou J, Rezvani Y, Gubbels MJ. Cell cycle-regulated ApiAP2s and parasite development: the Toxoplasma paradigm. **Current opinion in microbiology.** 2023 Dec 1;76:102383. PMID: 37898053.
- Wei G, Li S, **Ye S**, Wang Z, Zarringhalam K, He J, Wang W, Shao Z. High-resolution small RNAs landscape provides insights into alkane adaptation in the marine alkane-degrader Alcanivorax dieselolei B-5. **International Journal of Molecular Sciences**. 2022 Dec 15;23(24):15995. PMID: 37057895.
- Zheng Y, Wang D, **Ye S**, Chen W, Li G, Xu Z, Bai S, Zhao F. Auxin guides germ-cell specification in Arabidopsis anthers. **Proceedings of the National Academy of Sciences**. 2021 Jun 1;118(22): e2101492118. PMID: 37898053.

<u>In preparation</u>

- **Ye S#**, Dass S#, Elsworth B#, Paul AS, Zarringhalam K, Duraisingh MT. Uncovering the essentialome of *Babesia divergens* by piggyBac transposon mutagenesis. (#co-first author)
- Kumar M#, **Ye S#**, Diffendall G#, Zarringhalam K, Duraisingh MT. Lactylation of epigenetic regulators controls gene expression linked to sexual commitment in *Plasmodium falciparum*. (#co-first author)
- Mann A#, **Ye S#**, Sullivan M,Zarringhalam K, Duraisingh MT. Inducible transposon mutagenesis in *Plasmodium falciparum* reveals the essential genes of IPP bypass in Apicomplast. (#co-first author)
- Zingl FG#, **Ye S#**, Balazs GI, Basta DW, Waldor MK. Spatial-temporal MultiCAST *in vivo* screens in gastrointestinal tract of mouse identifies the dynamic requirements for the colonization of *Vibrio Cholerae*. (#co-first author)

PRESENTATIONS

- Sida Ye et al. “The essential genome of *Plasmodium knowlesi* reveals determinants of antimalarial susceptibility”. Invited online talk in Zhejiang University, China, Dec 06th
- Sida Ye et al. “The essential genome of *Plasmodium knowlesi* reveals determinants of antimalarial susceptibility”. Invited talk in Chinese Genomics Meet-up online 2025, Oct 22th
- Sida Ye et al. “Uncovering the essentialome of *Babesia divergens* by piggyBac transposon mutagenesis”. Poster presented at: MOLECULAR PARASITOLOGY MEETING 2023, Sep 14-18
- Sida Ye et al. “Single-cell transcriptomic analysis of cold-induced quiescence in *Babesia divergens*”. Poster presented at: MOLECULAR PARASITOLOGY MEETING 2022, Sep 14-18

REVIEWER EXPERIENCES

BMC Infectious Diseases (2)
Invited reviewer for Gen^2 2026 | ICLR 2026

AWARDS&HONORS

UMB Dean’s Doctoral Research Fellowship (2025)
Founder Scholarship (Peking University 2019)
First-class Scholarship for Graduate Students (Peking University 2018)
Second-class Scholarship for Graduate Students (Peking University 2017)
The Top2 Distinguished Poster Award in "The Academic Week” (Peking University 2016)
Lv Yichang Scholarship (Peking University 2015)
Scholarship for Outstanding Students (Huazhong Agricultural University 2010-2014)
National Encouragement Scholarship from Ministry of Education in China (2011)

RESEARCH STATEMENTS

Before my Ph.D., I received six years of intensive wet-lab training at Peking University (China) and the University of Cambridge (U.K.), where I investigated the molecular mechanisms regulating stamen morphogenesis in *Oryza sativa* and *Arabidopsis thaliana* (PBJ, PMID: 37057895; PNAS, PMID: 34031248).During my four years of doctoral training at University of Massachusetts Boston and the Harvard T.H. Chan School of Public Health, I transitioned into computational biology, receiving formal training in mathematical and machine learning modeling in Prof. Kourosh Zarringhalam's group (Department of Mathematics, UMass Boston) and continued experimental training in microbiology in Prof. Manoj T. Duraisingh's laboratory (Department of Immunology and Infectious Diseases). My Ph.D. research focused on developing mathematical models and bioinformatics pipelines to quantify gene essentiality and fitness at genome scale using transposon insertion mutagenesis data. I have applied these approaches to *Plasmodium knowlesi* (Science, PMID: 39913579), as well as *Plasmodium falciparum* and *Babesia divergens* (manuscripts in preparation). Currently, I am an HHMI-employed postdoctoral scientist in Prof. Matthew K. Waldor's laboratory at Harvard Medical School and Brigham and Women's Hospital.

Contributions

Essential Genome of *Plasmodium knowlesi*: I led the computational analysis, mathematical modelling and webapp development in this project. Specifically, I developed the improved TN-Seq/QI-Seq analysis pipeline and several mathematical models (HMS, OIS, FIS, truncation model) to extract transposon insertion patterns, identifying a *Plasmodium knowlesi*-specific essential metabolic pathway. Notably, we found that farnesyl pyrophosphate synthase (FPP/GGPPS), a proposed and widely accepted *P. falciparum* drug target, is non-essential in *P. knowlesi*. Using dihydroartemisinin (DHA) and GNF179 perturbation experiments along with a large library of mutants, I developed a perturbation analysis pipeline and models (edgeR-based, sites-level model, cv inverse model) to quantify the increase and decrease of transposon insertions within the coding sequence unbiasedly at genome scale. we discovered novel drug resistant/sensitizing genes. Furthermore, I developed a lncRNA and genome structural variants identification pipeline based on bulk RNA-Seq and DNA-Seq in *P.knowlesi* and found the essential lncRNAs in the blood stage of *P. knowlesi.* I also mentored a master student to develop an online interactive visualization platform: https://umbibio.math.umb.edu/PkEssenDB/. I created 95% of the figures visualizing the complex and multi-dimensional data in the Science paper.

Transposon Mutagenesis in *Babesia divergens*: I spent one year in the wet lab of Prof. Manoj Duraisingh in Harvard T.H. Chan School, constructing transposon mutant libraries in *B. divergens*, an important emerging Apicomplexan parasite infecting cattle and humans. I also made use of the transposon library to perform cold-shock and drug perturbations screens. Furthermore, I integrated the datasets from the perturbed transposon mutant screens with scRNA-Seq datasets to begin to reveal the molecular mechanism of *B.divergens* respond to cold shock(manuscript in preparation). In addition, I initiated an AI-based gene essentiality prediction model in the Kourosh group.

Inducible Transposon Mutagenesis in *Plasmodium falciparum*: In collaboration with Sean Prigge's group at Johns Hopkins University, I designed Transposon Mutagenesis Library 2.0, integrating transposase directly into the parasite genome, significantly improving transposition efficiency, I led the computational analysis for this project (manuscript in preparation).

Epigenetic Regulation via Lactylation in *Plasmodium falciparum*: I developed an automated analysis pipeline for CUT&RUN/CHIP-Seq on high-performance computing clusters and identified the link between lactylation and sexual commitment in *P.falciparum*. Based on my differential binding analysis, I developed new models to reveal how differential concentration of lactylation induce *P.falciparum* differentiation(manuscript prepared).

Spatial-temporal MultiCAST *in vivo* screens in gastrointestinal tract of mouse: I led the computational analysis of this project in Waldor lab. I designed the preliminary analysis pipeline integrated with MaGeCK MLE module for the Spatial-temporal MultiCAST(a RNA-guided transposon mutagenesis system) *in vivo* screens, used Dynamic Time Warping clustering, co-fitness analysis and graphical guassian model to characterize the dynamic requirements for the colonization of *Vibrio Cholerae* in gastrointestinal tract(manuscript prepared).

Apart from above, I was also involved in several publications and contributed to small RNAs analysis, protein structure prediction and protein-ligand docking [IJMS, PMID: 36555635; PNAS, PMID: 38377214; COMICR, PMID: 37898053].